ECE 370/590 Laboratory Exercise 1: The Dipole Antenna

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[date]

1 Analytical Design (Exercise 1.1)

At $f = 1\,\text{GHz}$,

$$\lambda = \frac{c}{f} = \frac{2.998 \times 10^8\,\text{m/s}}{1 \times 10^9\,\text{Hz}} = 299.8\,\text{mm}, \qquad \frac{\lambda}{2} = 149.9\,\text{mm}. \tag{1}$$

Table 1: Initial design parameters.

Parameter	Symbol	Value
Total length (tip to tip)	L	150 mm
Conductor	—	Copper tape on foam board
Strip width	w	25.4 mm
Tape thickness	t	0.05 mm
Feed gap	g	12.7 mm

Question 1. What initial dimensions would you select for a dipole resonant near 1 GHz and why?

Answer. *[Answer here.]*

Question 2. What input resistance do you expect at resonance, and how does this compare with the 50 Ω impedance of the VNA?

Answer. *[Answer here.]*

Question 3. How should the antenna impedance change as the length is increased or decreased?

Answer. *[Answer here.]*

Question 4. How might the conductor diameter or width affect resonance and bandwidth?

Answer. *[Answer here.]*

2 CST Simulation (Exercise 1.2)

2.1 Model setup

[Solver (time domain), frequency range 0.7 GHz to 1.3 GHz, open (add space) boundaries, discrete 50 Ω port across the feed gap, conductor material (copper vs. PEC), foam board included or not, mesh cell count and convergence check. State the simulated length L.]

2.2 Reflection coefficient and input impedance

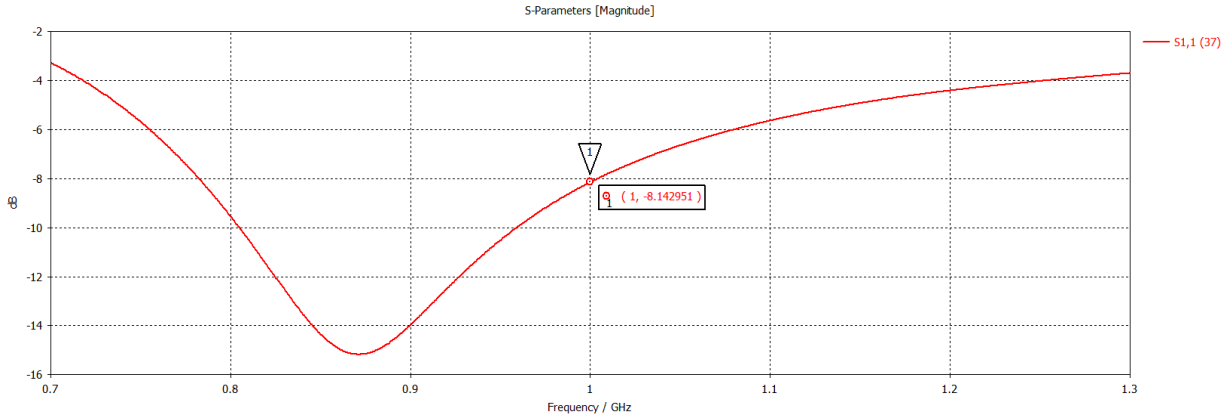


Figure 1: Simulated $|S_{11}|$ (dB) re 50 Ω, 1 in-wide tape dipole. Minimum −15.1 dB near 0.873 GHz; −8.14 dB at 1 GHz.

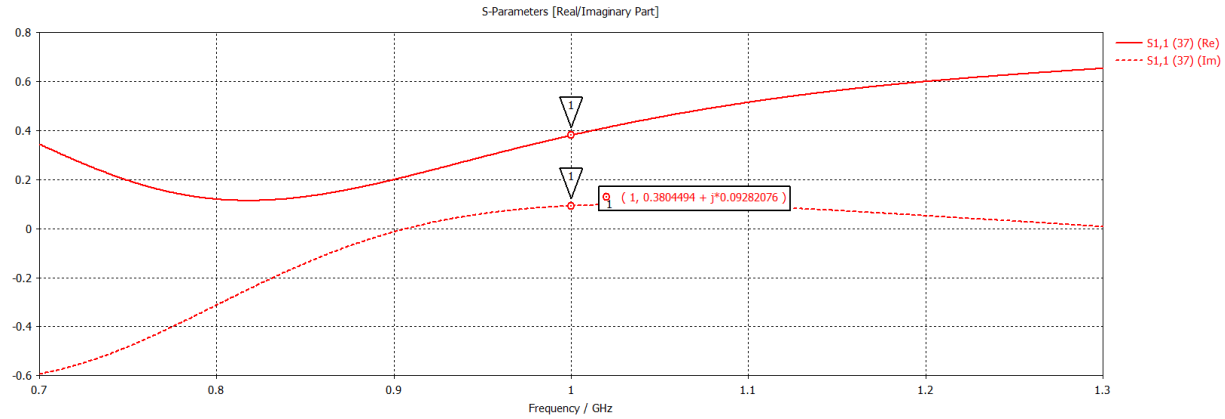


Figure 2: Simulated S_{11}, real and imaginary parts. At 1 GHz: $S_{11} = 0.380 + j0.093$.

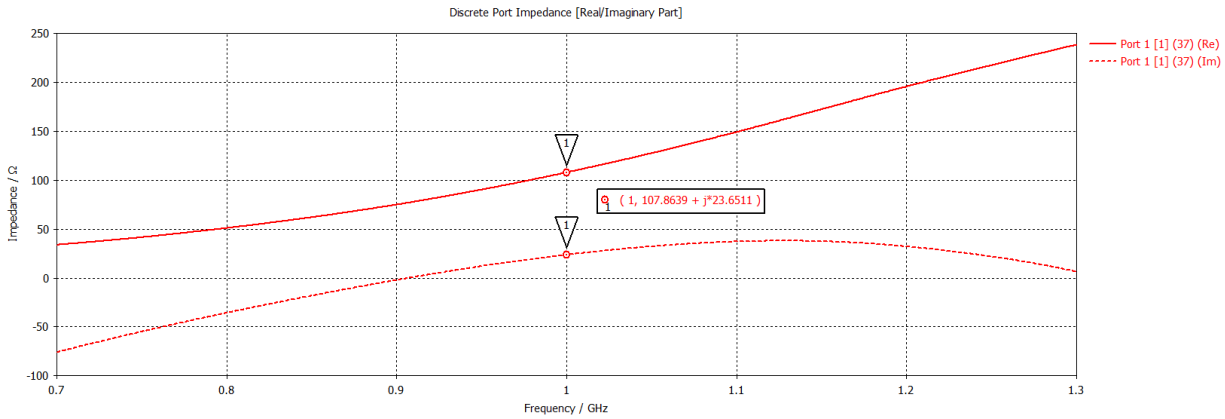


Figure 3: Simulated input impedance Z_{11}. At 1 GHz: $Z_{11} = 107.9 + j23.7\ \Omega$. Reactance crosses zero near *[≈0.90 GHz; read exact value from a marker]*.

Table 2: Simulated resonance summary (1 in tape, L = *[...]*).

Quantity	Value
Zero-reactance frequency $f_{X=0}$	*[≈0.90 GHz]*
R_{in} at $f_{X=0}$	*[≈75 Ω]*
Frequency of min $\lvert S_{11}\rvert$	*[0.873 GHz]*
Min $\lvert S_{11}\rvert$	−15.1 dB
−10 dB band	*[≈0.82 GHz to 0.93 GHz]*
Fractional bandwidth	*[...]*
Z_{11} at 1 GHz	$107.9 + j23.7\ \Omega$
Radiation efficiency at 1 GHz	0.9998
Directivity at 1 GHz	2.19 dBi

2.3 Surface current and far field

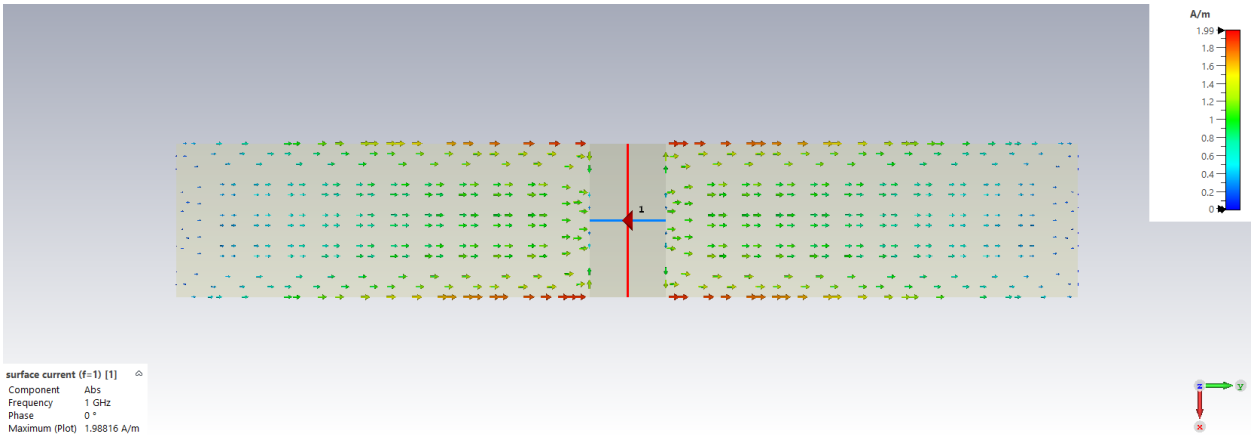
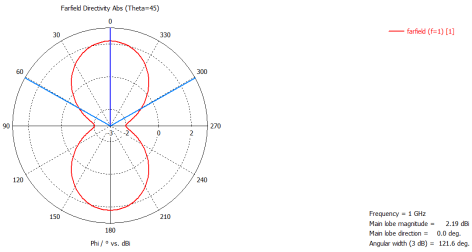
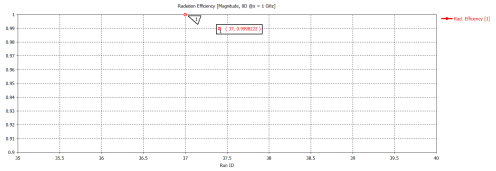


Figure 4: Simulated surface current magnitude at 1 GHz (max 1.99 A/m).



(a) Directivity cut at 1 GHz (main lobe 2.19 dBi, 3 dB width 121.6°).



(b) Radiation efficiency at 1 GHz: 0.9998.

Figure 5: Simulated far-field and efficiency.

2.4 Parameter sweep

[Length sweep for the 1 in model. Only the 0.5 in model currently has a sweep (`Plots/0.5 Inch/`, files `S1,1_130` to `S1,1_160`); either run the same sweep at 1 in or include the 0.5 in sweep and say so.]

> **Question 5.** How closely does the simulated resonant length agree with your analytical estimate?

Answer. *[Answer here.]*

> **Question 6.** What happens to the input impedance as the dipole length changes?

Answer. *[Answer here.]*

> **Question 7.** How does the predicted bandwidth depend on conductor thickness or width?

Answer. *[Answer here.]*

> **Question 8.** What current distribution is observed along the antenna near resonance?

Answer. *[Answer here.]*

Selected fabrication geometry: *[...]*

3 Fabrication (Exercise 1.3)

[Copper tape (1 in wide) on foam board; feed gap; SMA-to-screw-terminal adapter attachment; strain relief; how lengths were changed (trimmed one specimen from 18.5 cm downward, or separate specimens).]

Table 3: Design vs. as-built dimensions.

Dimension	Design	As built
Total length L	*[...]*	18.5 cm, 18 cm, 17 cm, 16 cm and 15 cm
Strip width w	25.4 mm	*[...]*
Feed gap g	*[...]*	*[...]*
Lead length	*[...]*	*[...]*

4 Calibration and Measurement (Exercise 1.4)

4.1 Calibration and reference plane

[LibreVNA SOL calibration over 0.5 GHz to 1.5 GHz (2 MHz steps, 501 points); reference plane at the end of the SMA test cable; adapter and leads not de-embedded (justify). Cable-movement diagnostic result.]

4.2 Measured S_{11}

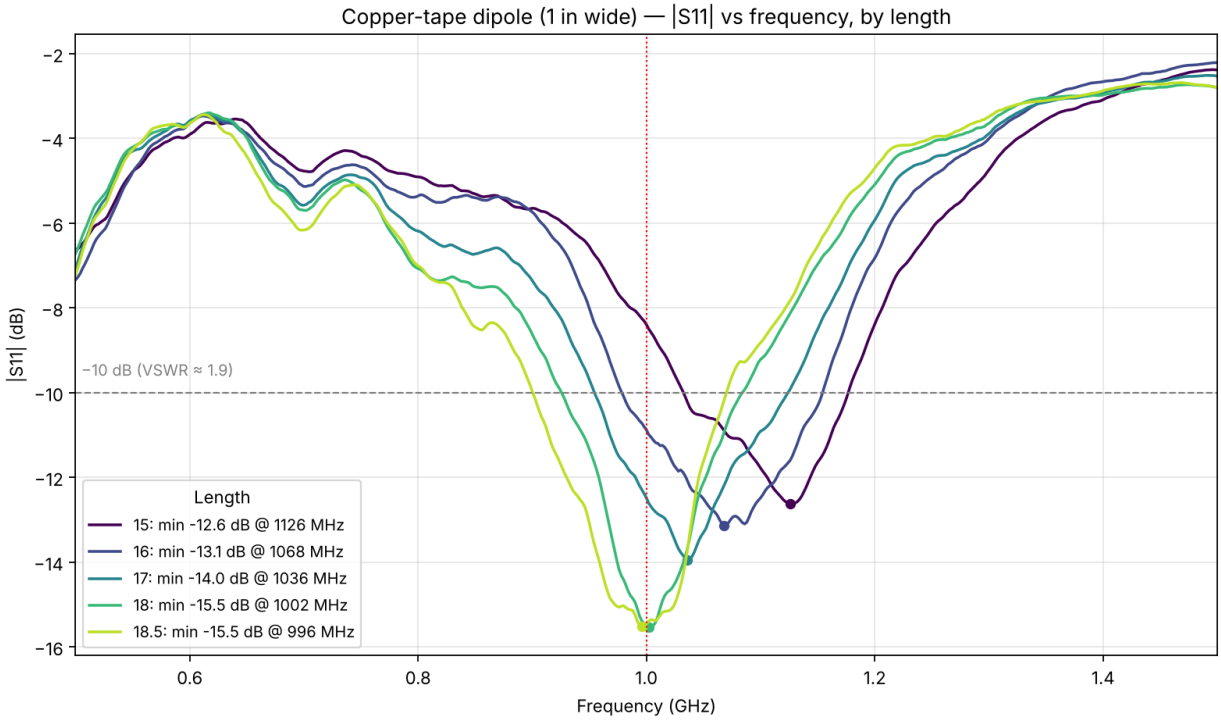
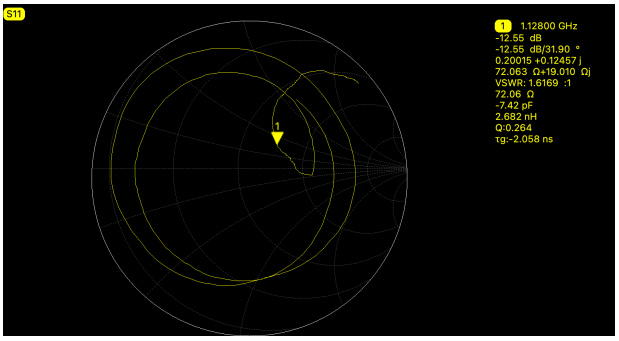
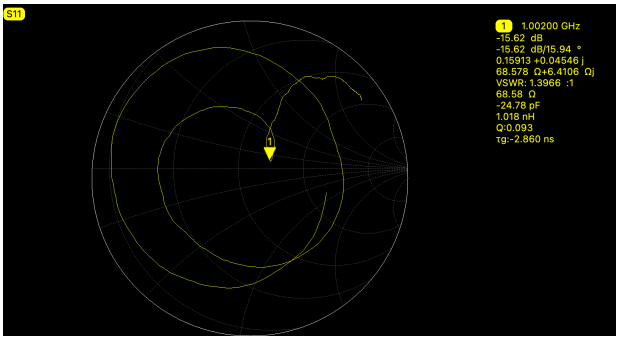


Figure 6: Measured $|S_{11}|$ of the 1 in copper-tape dipole for each length.



(a) Smith chart, $L = 15\,\text{cm}$.



(b) Smith chart, $L = 18\,\text{cm}$.

Figure 7: LibreVNA Smith charts at the shortest and near-resonant lengths. Marker at min $|S_{11}|$.

The input impedance at each marker follows from

$$Z_{\text{in}} = 50\,\frac{1 + S_{11}}{1 - S_{11}}\ \Omega. \tag{2}$$

Table 4: Measured results by length. Minimum $|S_{11}|$ and $-10\,\text{dB}$ band from the CSV exports; Z_{in} from the LibreVNA marker at (or near) the minimum.

L (cm)	$f_{\min\,\lvert S_{11}\rvert}$ (MHz)	min $\lvert S_{11}\rvert$ (dB)	$-10\,\text{dB}$ band (MHz)	BW (%)	Z_{in} at marker (Ω)
15	1126	−12.62	1034–1176	12.6	$72.1 + j19.0$ @ 1128
16	1068	−13.14	980–1154	16.3	$70.2 + j17.7$ @ 1088
17	1036	−13.95	956–1124	16.2	$72.0 + j12.2$ @ 1038
18	1002	−15.53	926–1082	15.6	$68.4 + j6.4$ @ 1002
18.5	996	−15.52	902–1070	16.9	$65.5 + j12.0$ @ 1000

[Zero-reactance frequency for each length. The CSVs hold magnitude only, so $X = 0$ cannot be found from them; re-export complex S_{11} (Touchstone `.s1p`) or read it from the Smith chart.]

5 Geometry Study (Exercise 1.5)

[Resonance shift vs. length, measured vs. CST. Compare slope $\Delta f/\Delta L$. Optional: quarter-wave (1 in vs. 0.5 in) result below.]

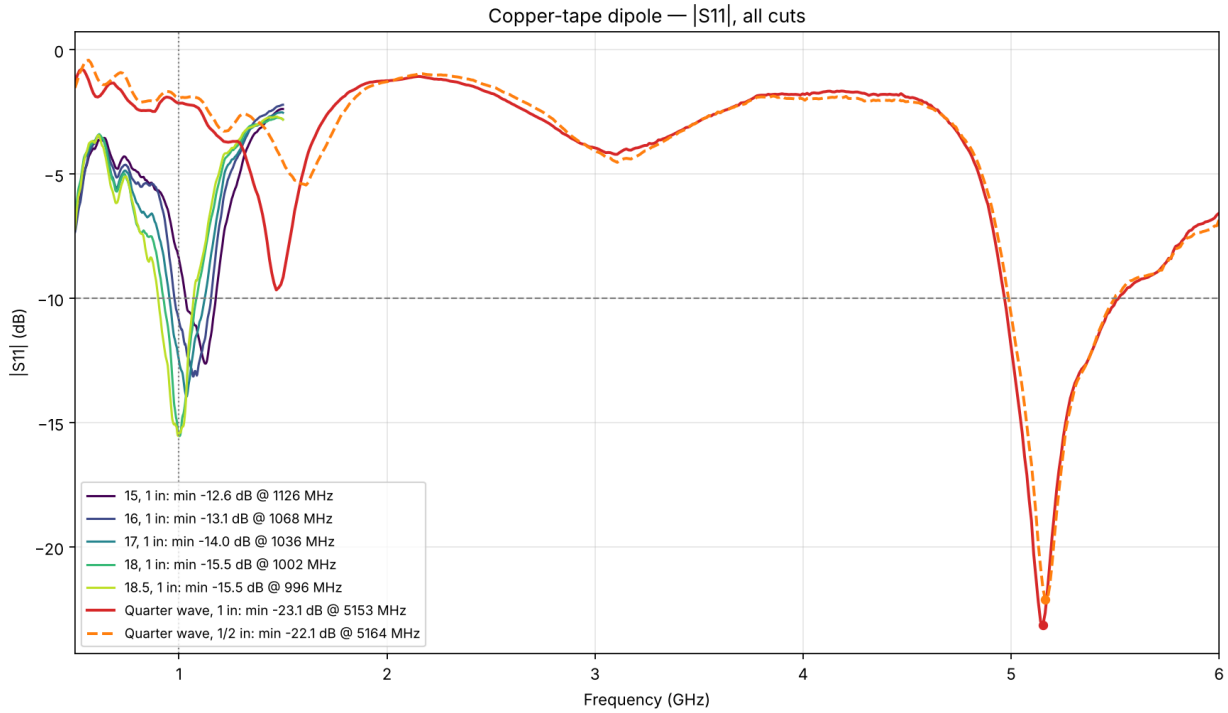


Figure 8: All measured cuts including the quarter-wave specimens (1 in and 0.5 in) to 6 GHz. *[Describe the quarter-wave configuration.]*

6 Comparison with Theory (Exercise 1.6)

[Analytical R_r from the thin-dipole current distribution at each measured length; compare with simulated and measured R_{in}.]

Table 5: Input resistance: theory, simulation, measurement.

L (cm)	R_r analytical (Ω)	R_{in} CST (Ω)	R_{in} measured (Ω)
15	*[...]*	*[...]*	72.1
16	*[...]*	*[...]*	70.2
17	*[...]*	*[...]*	72.0
18	*[...]*	*[...]*	68.4
18.5	*[...]*	*[...]*	65.5

Question 9. How does the analytical radiation resistance compare with the simulated input resistance?

Answer. *[Answer here.]*

Question 10. How closely does the measured resistance agree with simulation?

Answer. *[Answer here.]*

Question 11. What physical effects could account for differences?

Answer. *[Answer here.]*

Question 12. Can the measured input resistance alone establish the antenna's radiation efficiency? Explain.

Answer. *[Answer here.]*

7 Discussion

Resonant length vs. half wavelength

[Answer here.]

What determines impedance bandwidth

[Answer here.]

Copper tape vs. round wire

[Answer here.]

Feed-cable influence on a balanced antenna

[Answer here.]

VNA accuracy and de-embedding

[Answer here.]

Analytical theory vs. numerical simulation

[Answer here.]

8 Conclusion

[Answer here.]